

Task-1

1.Smtps

Desktop Computer

Convert ac power to stable dc voltage required by computer componenets.

LED/LCD Monitor

Provides efficient and regulated dc power to the display circuis.

Wi-Fi Router

Converts Household ac power and regulated dc power into the Low-voltage dc power needed by the router.

2.UPS

Desktop Computer

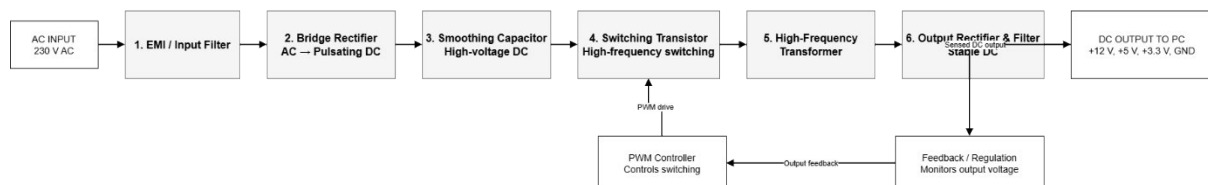
Provides temporary backup power during a power cut and prevent sudden shutdown.

CCTV System

Keep cameras and recording equipment running during power interruptions.

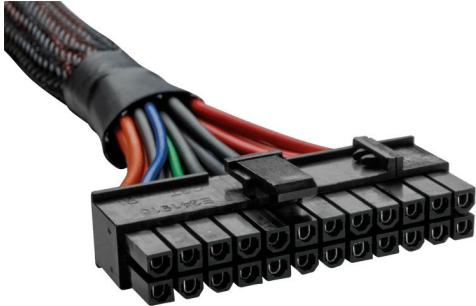
Task-2

Typical Desktop PC SMPS – Internal Components



Task- 3

1.ATX 24-pin Motherboard Connector



Main Power for the motherboard.

Turn off and unplug the pc before connecting or disconnecting.

Align the clip correctly and press firmly until it locks.

2.Sata Power Connector



HDDs, SDDs and DVD

Make sure the L-shape connector is aligned correctly.

Never force it into the socket.

3.Molex Connector (4-pin)



Older Hard Drives, optical drives , cases fans and other peripherals.

Hold the plastic connector, not the wires, when disconnecting.

Put it straight out without excessive force.

4.PCie 6/8-pin Connector



Provides additional power to graphics cards (GPUs).

Common versions are 6-pins, 8-pins and 6+2 pins.

Use only the PCIe cable from the PSU and make sure the connector is fully seated before powering on.

Task-4

Steps to connect SMPS to a pc

Connect the motherboard – Plug the 24-pin ATX power cable from SMPS into the motherboard's 24-pin power socket.

Connect CPU power – Connect the 4+4-pin (8pin) EPS/CPU power cable to the CPU socket near the processor.

Connect the graphics card (GPU) - Use the required PCIe 6-pin, 8-pin (6+2), or 12v- 2x6/16 pin cable, depending on the graphics card.

Connect Storage Devices – Use SATA power connectors from SMPS for SATA SSDs and HDDs. A SATA data cable separately connects each drive to the motherboard. M.2 NVMe SSDs do not need separate SMPS power cable.